

Project Installation

Unity 2022.3.41d1

Download the project from git

(I think the required packages will be also installed automatically)

Works with Oculus Quest 3

Project structure

Most of this work's original files (i.e. Scripts, prefabs, customized UMA Assets) are in the "Avatar creator" Assets folder

In the "Avatar creator/Scenes" folder, the scene called "**Customized Options**" is the actual main scene of this work. Other ones were used for some UMA testing.

How to properly uninstall the APK from Oculus Quest

While consequently building apps for Oculus, sometimes a problem may happen, that the Apk is not getting deployed. You can try deleting an App from the VR headset itself - but in reality, it won't be deleted and will be preventing further deployments:

<https://discussions.unity.com/t/after-build-run-nothing-is-installed-on-my-quest/917901/7>

So the only way to clean uninstall the APK is by using adb.exe in the command prompt on a PC connected to the Oculus Headset:

```
adb uninstall <package name>
```

adb.exe can be found in the Android SDK that Unity uses.

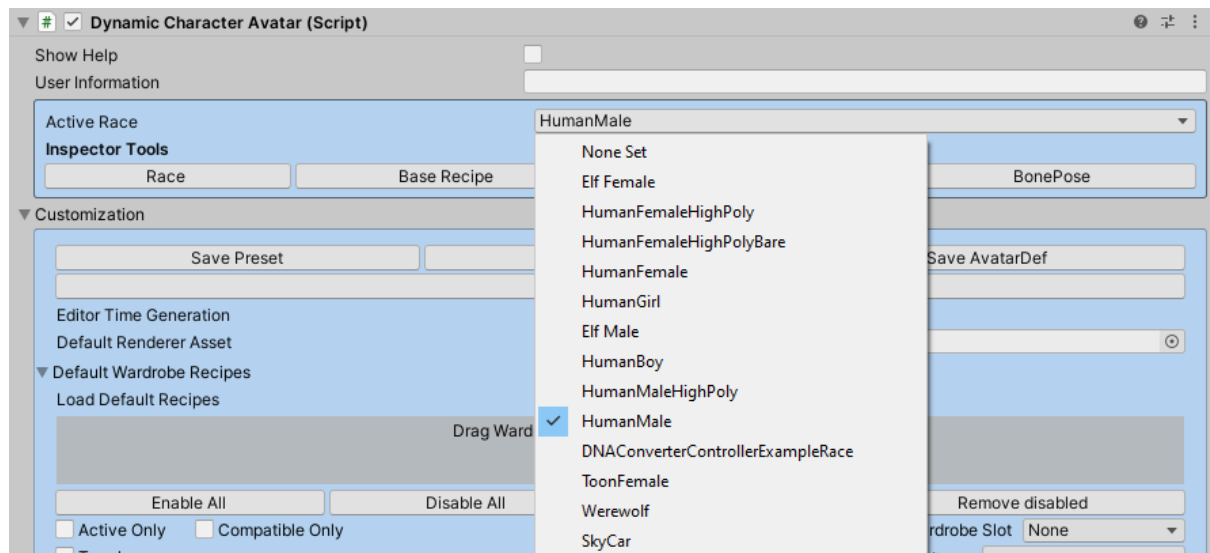
<Package name> can be found and changed in the project settings in Unity.

Unity and UMA

The character system in UMA requires multiple components in a scene, with the Dynamic Character Avatar being the most crucial for the development. It is attached to an actual human-representation prefab, and has most of the data in it: uma-race, wardrobe recipes, shared parameters, dna-values and many more.

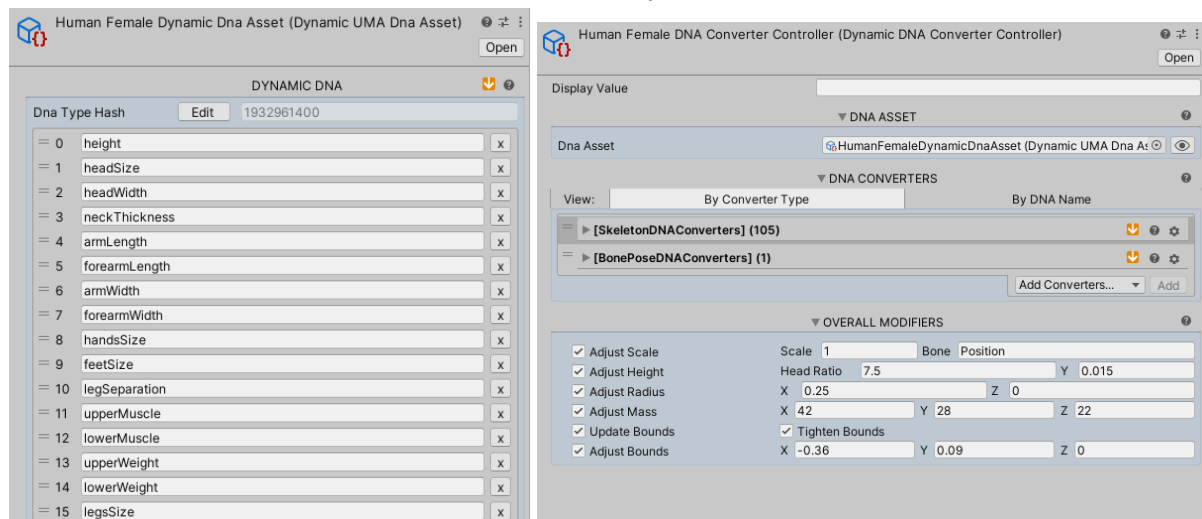
Standard content which comes with UMA includes male and female representation and defined settings and wardrobe recipes. More of that below.

For representing a different gender, UMA standard content has different base-models, and also different sets of dna-settings and wardrobe recipes. A set of base-models with dna-settings is called a "race"

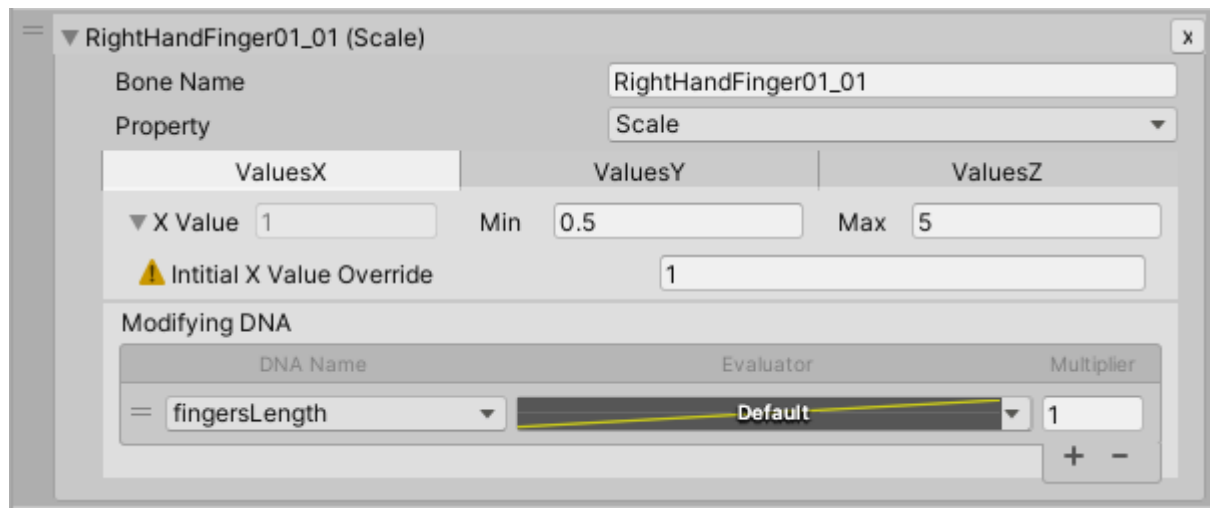


At first I made an in-Game UI drop down to make it possible to switch a character's gender between male, female and nonbinary, which would use HumanMale and HumanFemale races respectively. There haven't been any nonbinary-race built in so I decided to leave it as one of the other races until I make some progress with other ones (not in the final version)

The next step I tried was to access customizing options, already implemented in the standard content. Most of the body-changes, deformations and colors are controlled by dna-values. DNA-Value are defined in a Dynamic UMA Dna Assets and their affection on the bones or other characteristics of the model can be defined in a Dynamic Dna Converter Controller for each race



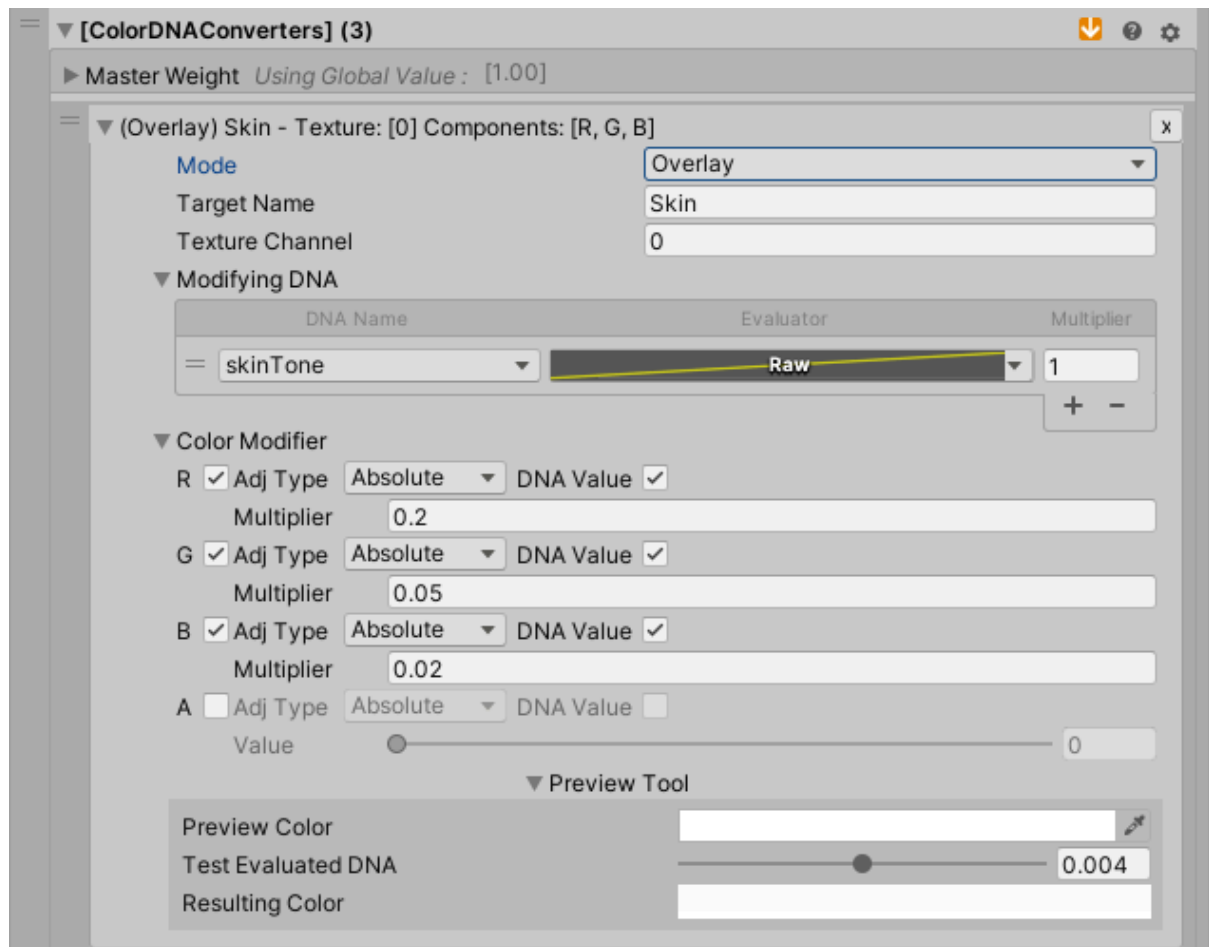
Now, an UMA-character game object consists of a hierarchy of bones and an UMA-controlled mesh renderer. DNA allows, for example, to define how a bone can be changed. For example, if a character should have a length-changeable fingers, DNA of fingersLength can control the X-Values of a scale of the beginning bone of each fingers. Some tweaking is needed of course with the min and max values.



That might be a way of adding older-people specific changes to a character. For example, I tried to create my own DNA parameter to change the posture of the character. It affects the spine bones rotation and shoulder bones positions

So the process of it looks like the following: defining a DNA in the main asset, finding the bones that should be affected by that DNA, creating a converter for each of them and setting up the modifiers.

In the case of colors, let's say changing the skin tone, some tweaking between rgb values is required. Setting the Target Overlay/Shared color name and texture channel. Uma finds it by its name.

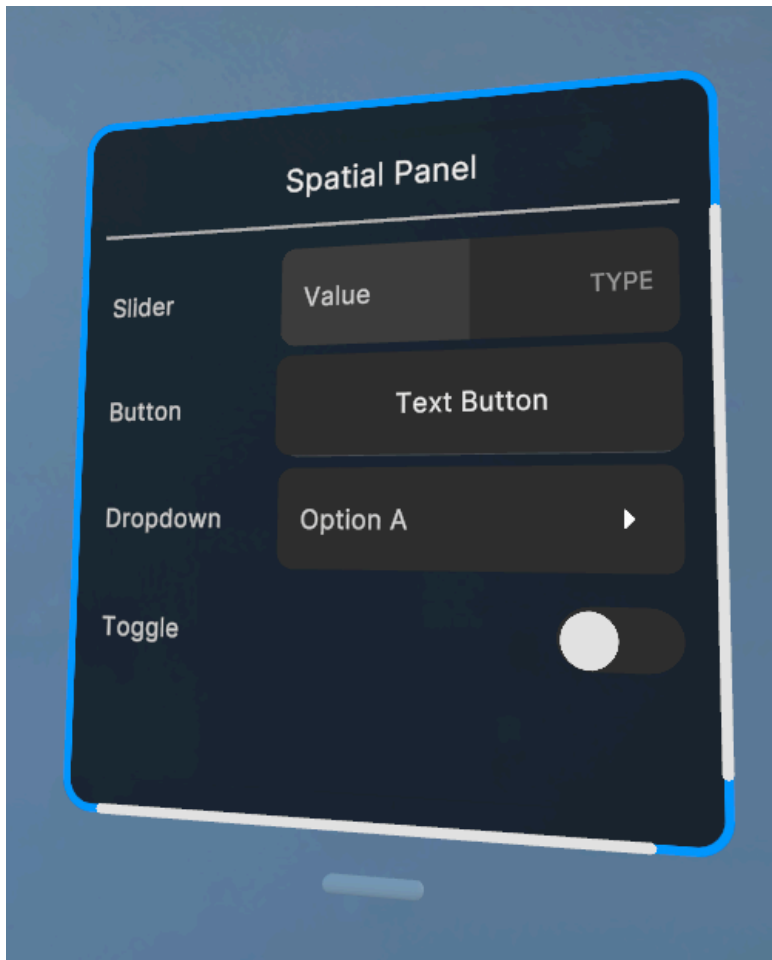


IN order to save time and develop as broadly as possible, I used Unity VRTemplateAssets as a base for my VR-controls and inGUI.

It has nice vr controller pointer-interaction included. Multi-input interactions for the elements as well: A slider can be controlled via Mouse cursor through the screen, as well as through pointer-interaction, as well as just poking with the controller. The whole panel is scrollable and can be thus scrolled with both a mouse wheel and a joystick on a controller, which is pointing at the scroll view.

The panel can also be grabbed, moved and scaled with controllers.

This panel was thus taken for a base for all menus in the app.



Humanoid Avatar Creator

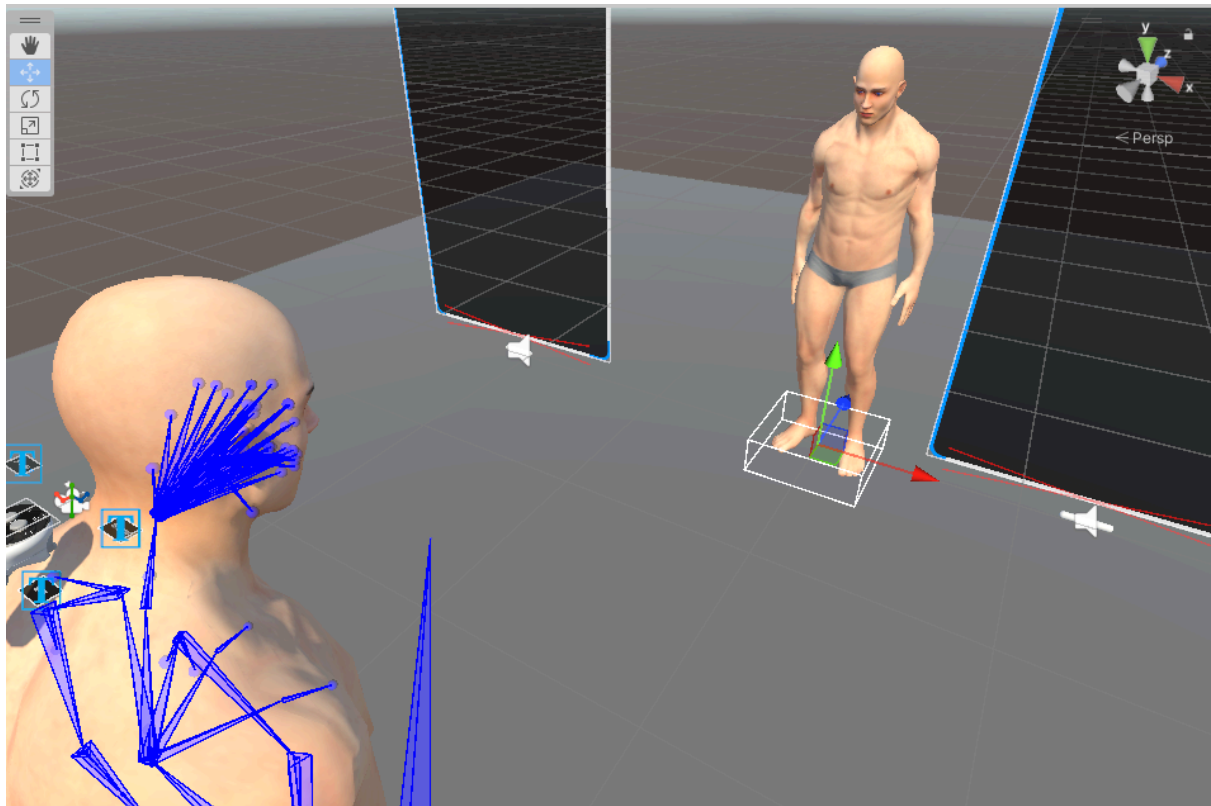
Although it is possible to choose a different entity type, the focus of this research lays upon the humanoid avatar creation. So the majority of the changes are controlled by such a submodule. While the inGUI consists of multiple spatial panels from above, the Humanoid Avatar Creator makes use of the Character Menu and Menu Settings. The humanoid Avatar Creator has 5 States: Skin, Body, Head, Hair and Outfit. Each of this states fills Menu Settings with specific controls, consisting of needed controlled elements such as sliders, toggles and buttons

Dynamic Character Avatar in the Scene

Initially there was only one UMA character controller to test out the possibilities of the changes of different DNAs and viewing changes. With the addition of VR-interface to the project I started working on the virtual body, which is per design is what the user actually is working on. This virtual body is seen from the user's perspective and is controlled by the user through body movements via controllers.

So this virtual body was the second UMA character controller(called Virtual Body Avatar). Now, with 2 UMAs I had to synchronize the changes made through the avatar creator UI. For that I created a special Humanoid Avatar Manager which encapsulates the possible changes of Dynamic Character Avatar in a way that the changes are being done for the both objects at the same time.

The latter setup actually has three UMA characters under this structure simultaneously(see below).



(early stage with 2 UMAs: Virtual Body Avatar and one for viewing changes)

The next important task was to figure out how the wardrobe recipes work and what they actually are.

Each UMA race has wardrobe slots - a list of slots (Body, Face, Hair, Hands etc). An UMA Slot recipe is a content for any of such a wardrobe slot, with any data that might be used: textures, meshes (UMA Documentation reference).



(two characters with different slot recipes for clothing(Torso and Legs wardrobe slots) and different hair slot recipes(Hair wardrobe slots))

Those slot recipes may be added to a DCA and stacked. Some may work as overlays, some as additional meshes, like hair or some volumetric clothes, and may be used for many purposes, like changing clothes, physique, skin texture and hiding other slots or body parts. Slot recipes might be set up to be shared between UMA-races or to be shown only on specific UMA-races.

One unconventional use for slot recipes I did was to hide the head meshes for the Virtual Body Avatar, since the Camera shouldn't see their insides, but the hands and the rest of the body. For that I created two Slot recipes "No Head" and "No Hair" and applied it selectively for the Virtual Body Avatar.

Uma had already included some sets of clothes included as a standard content. In research we tried to expand it, although not excessively, as the process of finding the 3d models and converting them into proper slots has proven to be quite time consuming.

Known problems

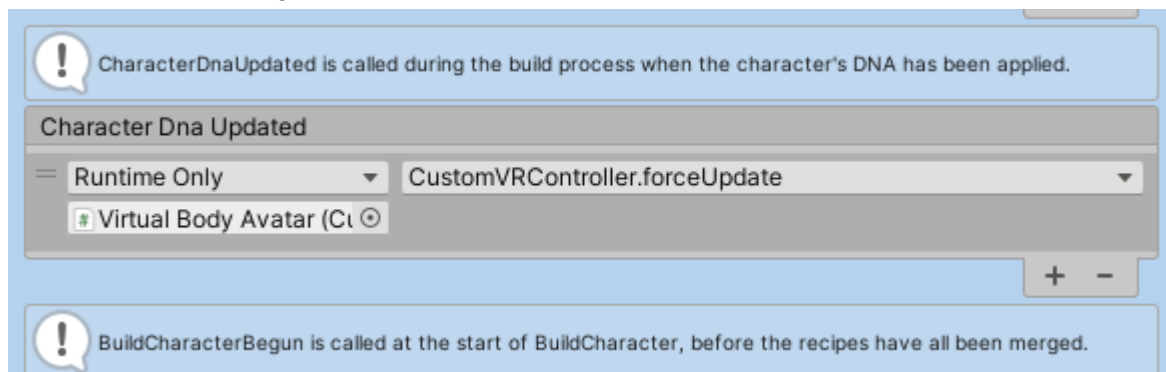
AKA The Main Bug

DCA has many avatar settings, like “Raw Avatar”, or “Keep Avatar”, and the animator, and it’s “character avatar” influence the bones of the UMA. As well as the Rigging does. So for example, if Raw Avatar is set, the changes of DNA are happening smoothly, but they are not all functioning, because some of them are being locked(or not defined, I have no clue) by the animator or it’s lacking. On the other hand, keeping an avatar without resetting it for a different race, makes the face of a newly selected race all queer and broken. No animator at all doesn’t work, because Rigging, which is needed for VR controls, requires animator. So there are UMA + Animator + Avatar + Rigging...

Up until now I had a bug that each time something in the character is changed, it jiggles for a frame or so in a weird pose.

Turns out each time UMA recreates a character, resets its rigging to a default state(initial positions before the scene started).

Setting this in DCA events makes those IK targets simultaneously get back to the proper position without losing a frame:



Helpful Links:

<https://www.youtube.com/watch?v=W0mK2k-lbGE&list=PLkDHFObfS19yYJoQsMR0dCGjCLsrF1dB4>

- a bit outdated, but still useful insight in DNAs, Race and most Data Assets in UMA

https://www.youtube.com/watch?v=9DsHxCdJSRo&list=PLkDHFObfS19yOjGzDUqvoq-oTy97g2G_J

- content creation for UMA